

UNIVERSIDADE ESTADUAL DE MONTES CLAROS – UNIMONTES
CENTRO DE CIÊNCIAS EXATAS E TECNOLÓGICAS – CCET
DEPARTAMENTO DE CIÊNCIAS DA COMPUTAÇÃO – DCC

ATIVIDADE AVALIATIVA II

RAMON LOPES DE QUEIROZ

MONTES CLAROS – MG

ABRIL/2026

RAMON LOPES DE QUEIROZ

ATIVIDADE AVALIATIVA II

Atividade avaliativa apresentada para atendimento de requisito parcial para aprovação na disciplina Geometria Analítica e Álgebra Linear do Curso de Graduação em Bacharelado em Sistemas de Informação – 2º período

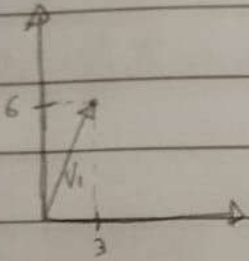
Professor: Antônio Wilson Vieira

MONTES CLAROS – MG

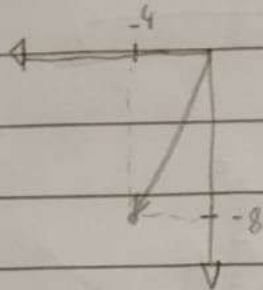
ABRIL/2026

Exercício - Decisão 3.1 - 3, 5, 7, 9, 13, 26, 31 -

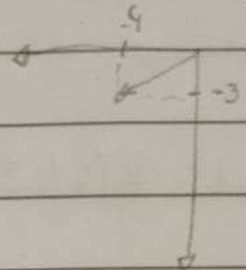
3-a. $V_1 = (3, 6)$



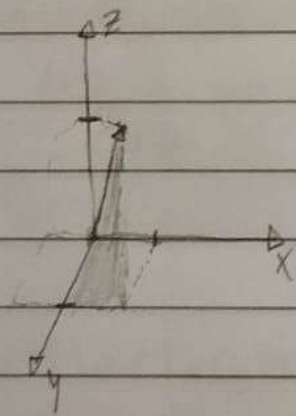
b. $V_2 = (-4, -8)$



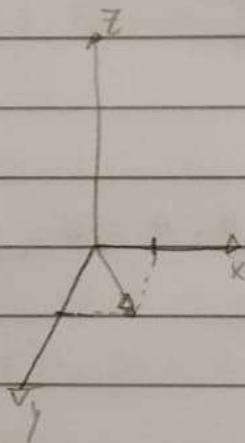
c. $V_3 = (-4, -3)$



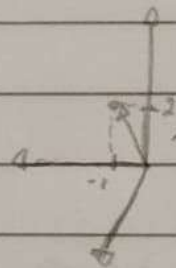
d. $V_4 = (3, 4, 5)$



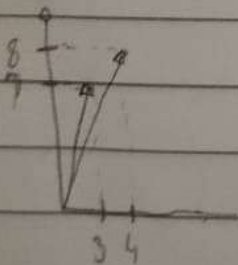
e. $V_5 = (3, 3, 0)$



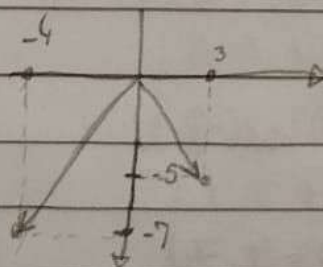
f. $V_6 = (-1, 0, 2)$



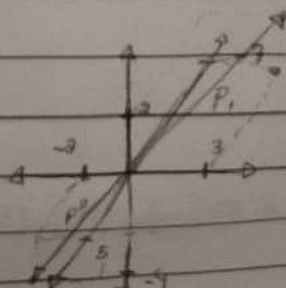
5-a. $P_1(4, 8), P_2(3, 7)$



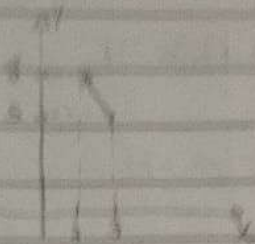
b. $P_1(3, -5), P_2(-4, -7)$



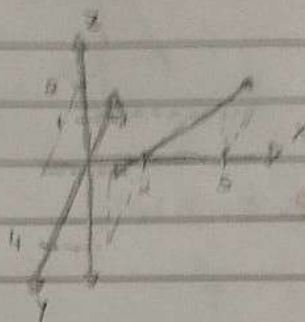
c. $P_1(3, -7, 2), P_2(-2, 5, -4)$



7-a. $P_1(1,5), P_2(2,2)$



8. $P_1(5,-2,1), P_2(2,4,2)$



9a. $A(1,1), U(1,2)$

$$U = B - A \Rightarrow B = U + A$$

$$V = (1+1, 1+2) \Rightarrow V = (2,3)$$

10. $B(-1,-1,2), U(1,1,3)$

$$A = B - U$$

$$V = (-1-1, -1-1, 2-3) \Rightarrow V = (-2, -2, -1)$$

13-a. $u = (4,-1), v = (0,5), w = (-3,-3)$

$$u + w = (4,-1) + (-3,-3) = (1,-4)$$

b. $(0,5) - 3 \cdot (4,-1)$
 $(-12,8)$

c. $2u - 10w = 2(4,-1) - 10(-3,-3) = (8,-2) - (-30,-30) = (38,28)$

$$d. -3v - 2(u + 2w) = (4, -1) + 2(-3, -3) + 3v$$

$$3(0, 5) - 2(-2, -7)$$

$$(4, 29)$$

$$e. -3(w - 2u + v) = 3((-3, -3) - 2(4, -1) + (0, 5))$$

$$-3(-11, 4)$$

$$(33, -12)$$

$$f. (-2u - w) + 5(v + 3w)$$

$$-2(4, -1) - (0, 5) + 5((0, 5) + 3(-3, -3))$$

$$(-8, -3) + (45, -20)$$

$$(37, 17)$$

$$26. C_1(1, 2, 0) + C_2(2, 1, 1) + C_3(0, 3, 1) = (0, 0, 0)$$

$$\begin{cases} a + 2b + 0 = 0 & 1 \ 2 \ 0 \ 0 \\ 2a + b + 3c = 0 & 2 \ 1 \ 3 \ 0 \\ 0 + b + c = 0 & 0 \ 1 \ 1 \ 0 \end{cases}$$

$$C_1 = 0$$

Three linearly independent.

$$C_2 = 0$$

$$C_3 = 0$$

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$b = -c$$

$$a + 2(-c) = 0, \ 2c$$

$$2(-c) - c + 3c = 0$$

$$c = 0$$

1 1 1

31- $C_1(-2, 9, 6) + C_2(-3, 2, 1) + C_3(1, 7, 5) = (0, 5, 4)$

$$\begin{cases} -2a - 3b + c = 0 \\ 9a + 2b + 7c = 5 \\ 6a + b + 5c = 4 \end{cases} \Rightarrow \begin{bmatrix} -2 & -3 & 1 & 0 \\ 9 & 2 & 7 & 5 \\ 6 & 1 & 5 & 4 \end{bmatrix} \xrightarrow{\substack{-\frac{1}{2}L_1 \\ \frac{1}{2}L_1}} \begin{bmatrix} 1 & \frac{3}{2} & -\frac{1}{2} & 0 \\ 9 & 2 & 7 & 5 \\ 6 & 1 & 5 & 4 \end{bmatrix} \xrightarrow{\substack{L_2 - 9L_1 \\ L_3 - 6L_1}} \begin{bmatrix} 1 & \frac{3}{2} & -\frac{1}{2} & 0 \\ 0 & -\frac{23}{2} & \frac{23}{2} & 5 \\ 0 & -8 & 8 & 4 \end{bmatrix} \xrightarrow{-\frac{2}{23}L_2} \begin{bmatrix} 1 & \frac{3}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & -1 & -\frac{10}{23} \\ 0 & -8 & 8 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{3}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & -1 & -\frac{10}{23} \\ 0 & -8 & 8 & 4 \end{bmatrix} \xrightarrow{L_3 + 8L_2} \begin{bmatrix} 1 & \frac{3}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & -1 & -\frac{10}{23} \\ 0 & 0 & 0 & \frac{12}{23} \end{bmatrix}$$

SI $\rightarrow$ Não tem escalonamento possível.

Exercício - Resolução 3.2 (Ímpares) -

1-a. $V = (4, -3)$ $\|V\| = \sqrt{4^2 + (-3)^2} = \sqrt{25} = 5$

$$u = \frac{1}{\|V\|} \cdot V = \frac{1}{5} \cdot (4, -3) = \left(\frac{4}{5}, -\frac{3}{5}\right)$$

$$-u = \left(-\frac{4}{5}, \frac{3}{5}\right) = \left(\frac{-4}{5}, \frac{3}{5}\right)$$

b. $V = (2, 2, 2)$ $\|V\| = \sqrt{2^2 + 2^2 + 2^2} = \sqrt{12} \approx 3,46 = 2\sqrt{3}$

$$u = \frac{1}{2\sqrt{3}} (2, 2, 2) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$-u = \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$$

c. $V = (1, 0, 2, 1, 3)$ $\|V\| = \sqrt{1^2 + 0^2 + 2^2 + 1^2 + 3^2} = \sqrt{15}$

$$u = \left(\frac{1}{\sqrt{15}}, 0, \frac{2}{\sqrt{15}}, \frac{1}{\sqrt{15}}, \frac{3}{\sqrt{15}}\right)$$

$$-u = \left(-\frac{1}{\sqrt{15}}, 0, -\frac{2}{\sqrt{15}}, -\frac{1}{\sqrt{15}}, -\frac{3}{\sqrt{15}}\right)$$

3.a. $u = (2, -2, 3)$ $v = (1, -3, 4)$ $w = (3, 6, -4)$

$$u+v = (2, -2, 3) + (1, -3, 4) = (3, -5, 7)$$

$$\|u+v\| = \sqrt{3^2 + (-5)^2 + 7^2} = \sqrt{83}$$

d. $\|u\| + \|v\|$
 $\sqrt{2^2 + (-2)^2 + 3^2} + \sqrt{1^2 + (-3)^2 + 4^2}$
 $\sqrt{17} + \sqrt{26}$

c. $\| -2u + 2v \| = 2 \| -u + v \|$
 $2 \cdot \sqrt{(-1)^2 + (-1)^2 + 1^2}$
 $2\sqrt{3}$

$$d \quad \|3u - 5v + w\|$$

$$\|(4, 15, -15)\| = \sqrt{4^2 + 15^2 + (-15)^2} = \sqrt{466}$$

$$3 \cdot (2, -2, 3) - 5(1, -3, 4) + (3, 6, -4)$$

$$(6, -6, 9) - (5, -15, 20) + (3, 6, -4)$$

$$(4, 15, -15)$$

$$5-a. \quad u = (-2, -1, 4, 5) \quad v = (3, 1, -5, 7) \quad w = (-6, 2, 1, 1)$$

$$3 \cdot (-2, -1, 4, 5) - 5 \cdot (3, 1, -5, 7) + (-6, 2, 1, 1)$$

$$(-6, -3, 12, 15) - (15, 5, -25, 35) + (-6, 2, 1, 1)$$

$$(-27, -6, 38, -19)$$

$$\|(-27, -6, 38, -19)\| = \sqrt{(-27)^2 + (-6)^2 + (38)^2 + (-19)^2} = \sqrt{729 + 36 + 1444 + 361} = \sqrt{2570}$$

$$b. \quad \|3u\| - 5\|v\| + \|w\|$$

$$\|(-6, -3, 4, 5)\| - 5\|(3, 1, -5, 7)\| + \|(-6, 2, 1, 1)\|$$

$$\sqrt{86} - 5\sqrt{94} + \sqrt{42}$$

$$c. \quad \| -\|u\| \cdot v \|$$

$$\| -\|(-2, -1, 4, 5)\| \cdot (3, 1, -5, 7) \|$$

$$\| -(\sqrt{46}) \cdot (3, 1, -5, 7) \|$$

$$\| -3\sqrt{46}, -\sqrt{46}, 5\sqrt{46}, -7\sqrt{46} \|$$

$$\sqrt{(-3\sqrt{46})^2 + (-\sqrt{46})^2 + (5\sqrt{46})^2 + (-7\sqrt{46})^2}$$

$$\sqrt{414 + 46 + 1150 + 2254}$$

$$\sqrt{3864}$$

$$7. v = (-2, 3, 0, 6)$$

$$\|v\| = \sqrt{(-2)^2 + 3^2 + 0^2 + 6^2} = \sqrt{49} = 7$$

$$\|Kv\| = 5$$

$$K \cdot \|v\| = 5$$

$$K \cdot 7 = 5$$

$$K = \pm \frac{5}{7}$$

$$9.a. (3, 1, 4) \cdot (2, 2, -4)$$

$$(3 \cdot 2, 1 \cdot 2, 4 \cdot -4)$$

$$(6, 2, -16)$$

$$6 + 2 - 16$$

$$-8$$

$P_6 =$

produto

$$(3, 1, 4) \cdot (3, 1, 4)$$

$$(9, 1, 16)$$

$$9 + 1 + 16$$

$$26$$

$$(2, 2, -4) \cdot (2, 2, -4)$$

$$(4, 4, 16)$$

$$24$$

$$1e. (1, 1, 4, 6) \cdot (2, -2, 3, -2)$$

$$2 - 2 + 12 - 12$$

$$0$$

$$(1, 1, 4, 6) \cdot (1, 1, 4, 6)$$

$$1, 1, 16, 36$$

$$54$$

$$(2, -2, 3, -2) \cdot (2, -2, 3, -2)$$

$$4 + 4 + 9 + 4$$

$$21$$

$$11a. u = (3, 3, 3), v = (1, 0, 4)$$

$$\sqrt{(3-1)^2 + (3-0)^2 + (3-4)^2}$$

$$\sqrt{2^2 + 3^2 + (-1)^2}$$

$$\sqrt{14}$$

$$b. u = (0, -2, -1, 1), v = (-3, 2, 4, 4)$$

$$\sqrt{(0+3)^2 + (-2-2)^2 + (-1-4)^2 + (1-4)^2}$$

$$\sqrt{9 + 16 + 25 + 9}$$

$$\sqrt{59}$$

$$c. u = (3, -3, -2, 0, 3, 13, 5), v = (-4, 1, -1, 5, 0, -11, 4)$$

$$\sqrt{(3+4)^2 + (-3-1)^2 + (-2+1)^2 + (0-5)^2 + (3-0)^2 + (13+11)^2 + (5-4)^2}$$

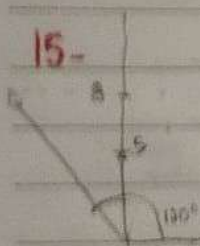
$$49 + 16 + 1 + 25 + 9 + 576 + 1$$

$$\sqrt{677}$$

$$13. a = \cos \theta = \frac{u \cdot v}{\|u\| \|v\|} = \frac{(3, 1) \cdot (3, 0) + (3, 4)}{\sqrt{9+1} \sqrt{9+0+6}} = \frac{15}{\sqrt{10} \sqrt{15}} = \frac{15}{3\sqrt{3} \cdot \sqrt{10}} \quad \text{Positivo} = \text{ângulo agudo}$$

$$b = \cos \theta = \frac{(0, -3) + (-2, 2) + (-1, 4) + (1, 4)}{\sqrt{0+4+1+1} \cdot \sqrt{9+4+16+16}} = \frac{-4}{\sqrt{6} \cdot \sqrt{45}} \quad \text{Negativo} = \text{Obtuso}$$

$$c = \cos \theta = \frac{(2, -4) + (-2, 1) + (-2, -1) + (0, 9) + (-3, 0) + (13, -11) + (5, 4)}{\sqrt{9+9+4+0+9+16+25} \cdot \sqrt{16+4+1+25+0+121+16}} = \frac{-136}{\sqrt{725} \cdot \sqrt{170}} \quad \text{Negativo} = \text{Obtuso}$$



$$|a| = 9, 120^\circ$$

$$120 - 90 = 30^\circ$$

$$|b| = 5, 90^\circ$$

$$9 \cdot 5 \cdot \cos(30^\circ)$$

$$\frac{45 \cdot \sqrt{3}}{2}$$

$$a \cdot b = 22,5\sqrt{3} \approx 38,97$$

17-a. $u \cdot (v \cdot w) = \text{Correto}$. $v \cdot w$ resulta em um escalar, e a multiplicação escalar se aplica entre dois vetores, e não entre escalar e vetor.

$$b. u \cdot (v + w) = \text{Correto}$$

c. $\|u \cdot v\| \neq \text{Não}$. Pois a norma é pra vetores, e a multiplicação resulta em um escalar.

$$d. (u \cdot v) \cdot w = \text{Correto}$$

$$19. a. (-4, -3) = \sqrt{(-4)^2 + (-3)^2} = \sqrt{25} = 5 \quad u = \left(\frac{-4}{5}, \frac{-3}{5} \right)$$

$$b. (1, 7) = \sqrt{1 + 49} = \sqrt{50} \quad u = \left(\frac{1}{\sqrt{50}}, \frac{7}{\sqrt{50}} \right)$$

$$c. (-3, 2, \sqrt{3}) = \sqrt{(-3)^2 + 2^2 + (\sqrt{3})^2} = \sqrt{16} = 4 \quad u = \left(\frac{-3}{4}, \frac{2}{4}, \frac{\sqrt{3}}{4} \right)$$

$$d. (1, 2, 3, 4, 5) = \sqrt{1^2 + 2^2 + 3^2 + 4^2 + 5^2} = \sqrt{55} \quad u = \left(\frac{1}{\sqrt{55}}, \frac{2}{\sqrt{55}}, \frac{3}{\sqrt{55}}, \frac{4}{\sqrt{55}}, \frac{5}{\sqrt{55}} \right)$$

$$21. u = \frac{v}{\|v\|} \quad w = m \cdot u$$

$$w = m \cdot \frac{v}{\|v\|}$$

$$23. a. u = (2, 3), v = (5, -7)$$

$$\frac{2 \cdot 5 + 3 \cdot (-7)}{\sqrt{2^2 + 3^2} \cdot \sqrt{5^2 + (-7)^2}} = \frac{-11}{\sqrt{13} \cdot \sqrt{74}} = \frac{-11}{\sqrt{962}}$$

$$b. \cos \theta = \frac{-6 \cdot 4 + (-2) \cdot 0}{\sqrt{36 + 4} \cdot \sqrt{16 + 0}} = \frac{-24}{\sqrt{40} \cdot \sqrt{16}} = \frac{-24}{4\sqrt{40}}$$

$$c. \cos \theta = \frac{1 \cdot 3 + 5 \cdot 3 + 4 \cdot 3}{\|u\| \cdot \|v\|} = 0 \quad \therefore \cos \theta = 0$$

$$d. \cos \theta = -2.1 + 2.7 \cdot 3 \cdot 4 = 0 \rightarrow \cos \theta = 0$$

$$25. a. u = (2, 2) \quad v = (4, -1)$$

$$|3 \cdot 4 + 2 \cdot (-1)| \leq \sqrt{3^2 + 2^2} \cdot \sqrt{4^2 + (-1)^2}$$

$$|10| \leq \sqrt{13} \cdot \sqrt{17}$$

✓

$$b. u = (-3, 1, 0), v = (2, -1, 3)$$

$$|-3 \cdot 2 + 1 \cdot (-1) + 0 \cdot 3| \leq \sqrt{3^2 + 1^2 + 0^2} \cdot \sqrt{2^2 + (-1)^2 + 3^2}$$

$$7 \leq \sqrt{10} \cdot \sqrt{14}$$

$$7 \leq \sqrt{140}$$

✓

$$c. u = (0, 2, 2, 1), v = (1, 1, 1, 1)$$

$$|0 \cdot 1 + 2 \cdot 1 + 2 \cdot 1 + 1 \cdot 1| \leq \sqrt{0^2 + 2^2 + 2^2 + 1^2} \cdot \sqrt{1^2 + 1^2 + 1^2 + 1^2}$$

$$5 \leq \sqrt{9} \cdot \sqrt{4}$$

$$5 \leq 6$$

✓

$$27. \|P - P_0\| = 1 \quad P - P_0 = (x - x_0, y - y_0, z - z_0)$$

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = 1^2$$

$$\text{Centro} = P_0 = (x_0, y_0, z_0)$$

$$S = \{ (x, y, z) \in \mathbb{R}^3 \mid (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = 1 \}$$

$$\text{Raio} = r = 1$$

$$29. \|V\| = \sqrt{V_1^2 + V_2^2 + \dots + V_n^2}$$

$$b. \|V\| = 0, \text{ se, e só se, } v_i = 0.$$

a.

$\|V\| \geq 0$, como a quadrado sempre
se é um +, a raiz é a soma todos,
então $\|V\| \geq 0$.

De $v_i \geq 0$, a soma sempre mais ou menos
que 0. O único caso é $V = 0$.

$$31.d. (u-v) \cdot w = u \cdot w - v \cdot w$$

$$(u \cdot w - v \cdot w) \rightarrow \text{Distributivität}$$

$$(u-v) \cdot w = \sum_{i=1}^n (u_i - v_i) w_i = \sum_{i=1}^n (u_i w_i - v_i w_i)$$

$$= \sum_{i=1}^n u_i w_i - \sum_{i=1}^n v_i w_i$$

$$32. a(u \cdot v) = u \cdot (a \cdot v)$$

$$a(u_1 v_1 + u_2 v_2 + \dots + u_n v_n)$$

$$a u_1 v_1 + a u_2 v_2 + \dots + a u_n v_n$$

$$u_1 (a v_1) + u_2 (a v_2) + \dots + u_n (a v_n)$$

$$33. \|u+v\| \leq \|u\| + \|v\|$$

$$(\|u+v\|)^2 = (\|u\| + \|v\|)^2$$

$$\|u\|^2 + 2(u \cdot v) + \|v\|^2 = \|u\|^2 + 2\|u\|\|v\| + \|v\|^2$$

$$2(u \cdot v) = 2\|u\|\|v\|$$

$$u \cdot v = \|u\|\|v\|$$

$$\|u\|\|v\| \cos \theta = \|u\|\|v\|$$

$$\cos \theta = 1$$

$$\theta = 0^\circ$$

Vektoren parallel und in die gleiche Richtung.